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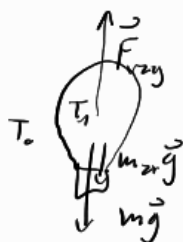
$$V = 450 \text{ m}^3$$

$$T_0 = 18^\circ\text{C}$$

$$p_0 = 1 \text{ bar}$$

$$m = 160 \text{ kg}$$

$$T_1 = ?$$



$$F_{vzg} = m_0g + m_{zv}g$$

$$\frac{p_0 VM}{RT_0} = m + \frac{p_0 VM}{RT_1}$$

$$\frac{p_0 VM - mRT_0}{RT_0} = \frac{p_0 VM}{RT_1}$$

$$T_1 = T_0 \frac{p_0 VM}{p_0 VM - mRT_0}$$

$$F_{vzg} = \int_0^{m_0} Vg$$

$$pV = \frac{m}{M} RT$$

$$m_0 = \frac{p_0 VM}{RT_1}$$

$$m_{zv} = \frac{p_0 VM}{RT_0}$$

$$T_1 = 140^\circ\text{C}$$

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$$S = 100 \text{ cm}^2$$

$$V_0 = 1 \text{ l}$$

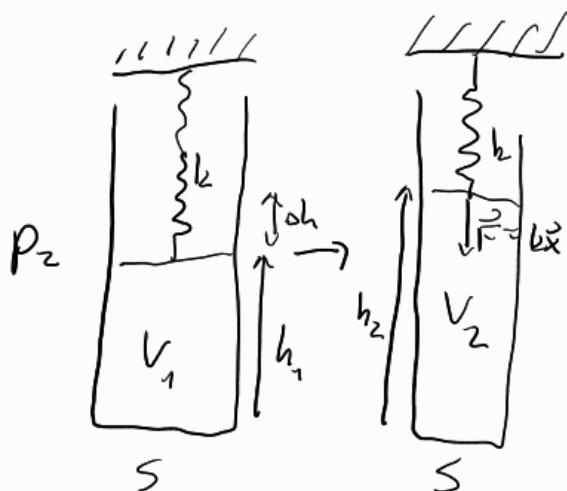
$$p_0 = 1 \text{ bar}$$

$$T_0 = 20^\circ\text{C}$$

$$k = 0,1 \frac{\text{N}}{\text{cm}}$$

$$T_1 = 80^\circ\text{C}$$

$$V_1 = ?$$



$$p_0 = \frac{kx_0}{S} + p_2$$

$$p_1 = \frac{k(x_0 + \Delta h)}{S} + p_2$$

$$p_1 - p_0 = \frac{k \Delta h}{S} \rightarrow p_1 = p_0 + \frac{k \Delta h}{S}$$

$$pV = \frac{m}{M} RT$$

$$V_1 = V_0 + S \Delta h$$

$$\Delta h = \frac{V_1 - V_0}{S}$$

$$\frac{p_0 V_0}{T_0} = \frac{p_1 V_1}{T_1}$$

$$\frac{T_1 p_0 V_0}{T_0} = \left(p_0 + \frac{k(V_1 - V_0)}{S} \right) V_1$$

$$p_0 V_1 + \frac{k}{S^2} V_1^2 - \frac{k}{S^2} V_1 V_0 - \frac{T_1}{T_0} p_0 V_0 = 0$$

$$V_1^2 \left(\frac{k}{S^2} \right) + V_1 \left(p_0 - \frac{k}{S^2} V_0 \right) - \frac{T_1}{T_0} p_0 V_0 = 0$$

$$V_1^2 + V_1 \left(\frac{p_0 S^2}{k} - V_0 \right) - \frac{T_1}{T_0} \frac{p_0 V_0 S^2}{k} = 0$$

$$V_1 = 1,2 \text{ l}$$

29/17

$T = \text{konst.}$
 $V = 10 \text{ cm}^3$

$p_0 = 1 \text{ bar}$

$t_1 = 1 \text{ min}$

$\phi_v = 0,2 \frac{\text{dm}^3}{\text{s}}$

$p_1 = ?$



$\phi_v \rightarrow \phi_m(t)$

$pV = \frac{m}{M} RT$

$\rho = \frac{pM}{RT}$

$\phi_m = \phi_v \rho = \phi_v \frac{pM}{RT} = \frac{dm}{dt}$

$m = m_0 - \int \phi_m dt$

$\frac{d}{dt}$

$\frac{dp}{dt} V = \frac{dm}{dt} \frac{RT}{M} = \phi_v p$

$\int_{p_0}^p \frac{dp}{p} = \int_0^t \frac{\phi_v}{V} dt$

$\ln \frac{p}{p_0} = -\frac{\phi_v}{V} t$

$p = p_0 e^{-\frac{\phi_v}{V} t}$

$p_1 = p_0 e^{-\frac{\phi_v}{V} t_1} = \underline{\underline{0,3 \text{ bar}}}$

kolokvij
 1986/87

(2.) nihanje

$S = 1 \text{ cm}^2$

$m = 400 \text{ g}$

t_{hor}

t_{ver}



b) ker hitro gibanje $\rightarrow T = \text{konst.}$ (adiabatsna spr.)

$pV^\gamma = \text{konst.}$

$F = ma = (p - p_0)S$

$V_0 = aS$

$V = (a - x)S$

$p = p_0 \left(\frac{V_0}{V} \right)^\gamma = p_0 \left(\frac{a}{a-x} \right)^\gamma$

$$-m\ddot{x} = p_0 S \left(\left(\frac{a}{a-x} \right)^k - 1 \right) \approx p_0 S k \frac{x}{a}$$

$x \ll a$:

$$\left(\frac{a}{a-x} \right)^k = \left(\frac{1}{1-\frac{x}{a}} \right)^k = \left(1 - \frac{x}{a} \right)^{-k} \approx 1 + k \frac{x}{a}$$

$$\ddot{x} = - \underbrace{\frac{p_0 S k}{m a}}_{\omega^2} x$$

$$t_{oh} = \frac{2\pi}{\omega} = \boxed{2\pi \sqrt{\frac{m a}{p_0 S k}}}$$

v)



$$-m\ddot{x} = p_0 S - mg \quad \xrightarrow{\text{nihanje okoli ravnovesja}} \text{novi } r$$

$a', p_0' \rightarrow$ v ravnovesju

$$(p_0 + p_0') S = mg$$

$$p_0' = \frac{mg}{S} + p_0$$

$$p_0 V_0 = p_0' V_0'$$

$$V_0' = \frac{p_0 V_0 S a}{p_0 + \frac{mg}{S}} = S a'$$

$$a' = a \frac{p_0 S}{p_0 S + mg} = a \left(\frac{1}{1 + \frac{mg}{p_0 S}} \right) \approx a \left(1 - \frac{mg}{p_0 S} \right)$$

$$\frac{t_{oh}}{t_{ov}} = \frac{\sqrt{\frac{a}{p_0}}}{\sqrt{\frac{a'}{p_0'}}} = \sqrt{\frac{a \cdot \left(\frac{mg}{S} + p_0 \right)}{p_0 a \left(1 + \frac{mg}{p_0 S} \right)^{-1}}} = \sqrt{\frac{\left(\frac{mg}{S} + p_0 \right) \left(1 + \frac{mg}{p_0 S} \right)}{p_0}} = \boxed{1 + \frac{mg}{p_0 S}}$$

12.1.6

$$\rho(h) = ?$$

$$T(h) = ?$$

$$p(h) = ?$$

a) izotermna - $T = \text{konst.}$

b) izentropna
↓

ne izmenjuje Q z
okolico (adiabatska
sprememba)

$$a) -dp = \rho g dh$$

$$-dp = \frac{\rho M}{RT} g dh$$

$$-\int_{p_0}^p \frac{dp}{\rho} = \int_0^h \frac{M}{RT} g dh$$

$$\ln \frac{p}{p_0} = - \frac{Mg}{RT} h$$

$$p = p_0 e^{-\frac{Mg}{RT} h} = \boxed{p_0 e^{-\frac{h}{h_0}}}$$

$$\rho = \frac{\rho M}{RT} = \underbrace{\frac{p_0 M}{RT}}_{\rho_0} e^{-\frac{h}{h_0}} = \boxed{\rho_0 e^{-\frac{h}{h_0}}}$$

$$b) -dp = \rho g dh$$

$$\int_{p_0}^p -p^{-\frac{1}{\kappa}} dp = \int_0^h \left(\frac{p_0}{p_0} \right)^{\frac{1}{\kappa}} g dh$$

$$\frac{p^{1-\frac{1}{\kappa}}}{1-\frac{1}{\kappa}} \Big|_{p_0}^p = - \int_0^h p_0^{-\frac{1}{\kappa}} g dh$$

$$\frac{\kappa}{\kappa-1} \left(p^{\frac{\kappa-1}{\kappa}} - p_0^{\frac{\kappa-1}{\kappa}} \right) = - \int_0^h p_0^{-\frac{1}{\kappa}} g dh$$

$$pV^{\kappa} = p_0 V_0^{\kappa}$$

$$p \left(\frac{V}{p} \right)^{\kappa} = p_0 \left(\frac{V_0}{p_0} \right)^{\kappa}$$

$$p = p_0 \left(\frac{p}{p_0} \right)^{\frac{1}{\kappa}}$$

$$p^{\frac{\kappa-1}{\kappa}} = \frac{\kappa-1}{\kappa} \cdot p_0^{\frac{\kappa-1}{\kappa}} \left(\frac{\kappa}{\kappa-1} - \int_0^h p_0^{-1} g dh \right) = p_0^{\frac{\kappa-1}{\kappa}} \left(1 - \frac{\int_0^h g dh (\kappa-1)}{p_0 \kappa} \right)$$

$$p = p_0 \left(1 - \frac{\int_0^h g dh (\kappa-1)}{p_0 \kappa} \right)^{\frac{\kappa}{\kappa-1}} = \boxed{p_0 \left(1 - \frac{h}{h_0} \frac{\kappa-1}{\kappa} \right)^{\frac{\kappa}{\kappa-1}}}$$

$$p = p_0 \left(\frac{r}{p_0} \right)^{\frac{1}{\alpha}} = \boxed{p_0 \left(1 - \frac{h}{h_0} \frac{x-1}{x} \right)^{\frac{1}{\alpha-1}}}$$

$$\frac{pM}{RT} = \frac{p_0 M}{RT_0} \left(\frac{r}{p_0} \right)^{\frac{1}{\alpha}}$$

$$T = T_0 \left(\frac{r}{p_0} \right)^{1-\frac{1}{\alpha}} = T_0 \left(\left(1 - \frac{h}{h_0} \frac{x-1}{x} \right)^{\frac{x}{x-1}} \right)^{\frac{x-1}{x}} = \boxed{T_0 \left(1 - \frac{h}{h_0} \frac{x-1}{x} \right)}$$

za oceno:

$$= T_0 - \frac{dT}{dh} \cdot h$$

$$\frac{T_0}{h_0} \frac{x-1}{x} = \frac{300K \cdot 0,4}{8320m \cdot 1,4} = 0,01 \frac{K}{m}$$

29/2

$$h = 20m$$

$$v = 3 m/s$$

$$c = 130 J/kgK$$

$$W_p - W_k = mc\Delta T$$

$$\frac{2gh - v^2}{2c} = \Delta T = 1,5^\circ C$$

$\Delta T = ?$

29/6



$$m_1 = 8t$$

$$W_{k2} - W_{k1} = m_2 c \Delta T$$

$$m_1 v_1 = 2m_2 v_2$$

$$m_2 = 8t$$

$$\frac{m_1 v_1^2}{2} - \frac{2m_2 v_2^2}{2} = m_2 c \Delta T$$

$$v_1 = \frac{v_2}{2}$$

$$v_2 = 10 m/s$$

$$\Delta T = \frac{m_1 (v_1^2 - \frac{v_2^2}{4})}{2m_2 c} = \frac{8 \cdot 10^3 kg \cdot \frac{10^2 m^2}{4}}{2 \cdot 10^3 kg \cdot 450 J/kgK} = 44,4 K$$

$$v_1 = 0$$

$$\Delta T = 7^\circ C \text{ odbijanja } m_1 = 10kg, c = 450 J/kgK$$

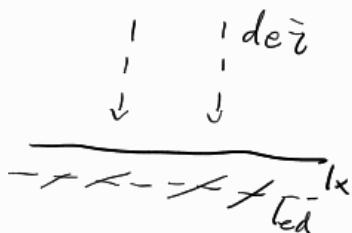
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$$T_{led} = 0^{\circ}C$$

$$\phi = 1 \frac{mm}{h}$$

$$T_{dež} = 6^{\circ}C$$

$$t = 12h$$



$$\rho_{led} = \frac{m_{led}}{S \cdot x}$$

$$m_{dež} = \rho_{dež} S \phi t$$

$$1,1 \cdot \rho_{led} = \rho_{voda}$$

$$x = ?$$

$$Q_{dež} = m_{dež} \Delta T = \rho_{voda} S \phi \Delta T t$$

$$Q_{led} = m_{led} q_t = \rho_{led} S x q_t$$

$$\rho_{voda} \phi t \Delta T_c = \rho_{led} x q_t$$

$$x = 1,1 \cdot \frac{\phi t \Delta T}{q_t} = 1mm$$

VPCIV OT zaradi
hitrosti dežja
je zanemarljiv

29/17

$$V = 40m^3$$

$$\eta = 0,65$$

$$\phi_m = 3 \frac{kg}{h}$$

$$q_s = 10,47 \frac{MJ}{kg}$$

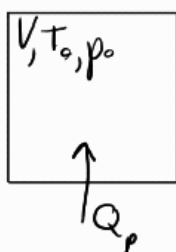
$$T_o = 15^{\circ}C$$

$$p_o = 1bar$$

$$t_n = 1min$$

$$\kappa = 1,4 = \frac{c_p}{c_v}$$

$$\Delta T_n = ?$$



$$Q_p = \phi_m \cdot t_n \cdot q_s \cdot \eta$$

$$Q_2 = m c_v \Delta T = \frac{p_o V / M}{T_o / R} c_v \Delta T = \frac{p_o V}{T_o} \frac{1}{\kappa - 1} \Delta T$$

$$c_p - c_v = \frac{R}{M}$$

$$\frac{M}{R} c_v = \frac{c_v}{c_p - c_v} = \frac{1}{\frac{c_p}{c_v} - 1} = \frac{1}{\kappa - 1}$$

$$\Delta T = (\kappa - 1) \phi_m t_n q_s \eta \frac{T_o}{p_o V} = 9,8K$$

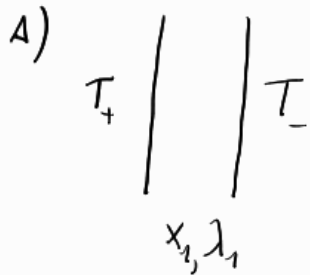
$$x_1 = 12 \text{ cm}$$

$$\lambda_1 = 1,1 \frac{\text{W}}{\text{mK}}$$

$$x_2 = 2 \text{ cm}$$

$$\lambda_2 = 0,05 \frac{\text{W}}{\text{mK}}$$

$$P = S \lambda \left(- \frac{dT}{dx} \right)$$



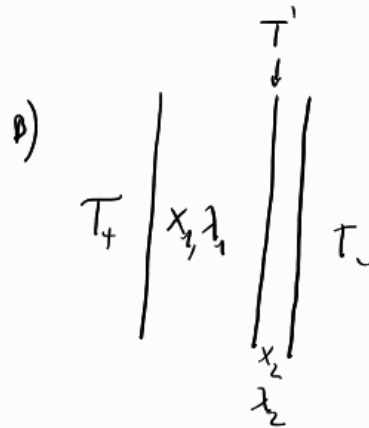
$$P = S \lambda_1 \frac{T_- - T_+}{x_1}$$

$$P_B = \left(\frac{\lambda_1 x_2 T_+ + \lambda_2 x_1 T_-}{\lambda_1 x_2 + \lambda_2 x_1} - T_+ \right) \frac{S \lambda_1}{x_1} =$$

$$= \frac{\lambda_2 x_1 (T_- - T_+)}{\lambda_1 x_2 + \lambda_2 x_1} \frac{S \lambda_1}{x_1}$$

$$\frac{P_B}{P_A} = \frac{\lambda_2 x_1 (T_- - T_+)}{\lambda_1 x_2 + \lambda_2 x_1} \frac{S \lambda_1}{x_1} \cdot \frac{x_1}{S \lambda_1 (T_- - T_+)} =$$

$$= \frac{\lambda_2 x_1}{\lambda_1 x_2 + \lambda_2 x_1} = 0,21$$



$$P_B = S \lambda_1 \frac{T' - T_+}{x_1} = S \lambda_2 \frac{T_- - T'}{x_2}$$

$$\lambda_1 x_2 (T' - T_+) = \lambda_2 x_1 (T_- - T')$$

$$T' (\lambda_1 x_2 + \lambda_2 x_1) = \lambda_1 x_2 T_+ + \lambda_2 x_1 T_-$$

$$T' = \frac{\lambda_1 x_2 T_+ + \lambda_2 x_1 T_-}{\lambda_1 x_2 + \lambda_2 x_1}$$

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$$2r = 30 \text{ cm}$$

$$x = 1 \text{ cm}$$

$$\lambda = 1 \frac{\text{W}}{\text{mK}}$$

$$T_i = 0^\circ \text{C}$$

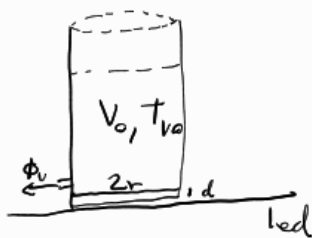
$$T_{v0} = 30^\circ \text{C}$$

$$V_0 = 20 \text{ l}$$

$$t_1 = 3 \text{ min}$$

$$\phi_v = 5 \text{ l/min}$$

$$T_{v1} = ?$$



$$P = S \lambda \frac{T_v - T_i}{x} = \frac{dQ}{dt}$$

$$dQ = m c dT_v = \rho (V_0 - \phi_v t) c dT_v$$

$$m = m_0 - \phi_v t = \rho (V_0 - \phi_v t)$$

$$- \frac{\rho (V_0 - \phi_v t) c dT_v}{dt} = \pi r^2 \lambda \frac{T_v - T_i}{x} dt$$

$$- \frac{\rho c x}{\pi r^2 \lambda} \int_{T_{v0}}^{T_{v1}} \frac{dT_v}{T_v - T_i} = \int_0^t dt$$

$$T_v - T_i = u$$

$$dT_v = du$$

$$V_0 - \phi_v t = s$$

$$- \frac{\rho c x}{\pi r^2 \lambda} \ln \frac{T_{v0} - T_i}{T_{v1} - T_i} = - \ln \frac{V_0}{V_0 - \phi_v t} = \ln \frac{V_0 - \phi_v t}{V_0} - dt = ds$$

$$-\ln \frac{T_{v0} - T_L}{T_{v1} - T_L} = \ln \left(\frac{V_0 - \phi_v t}{V_0} \right)^{\frac{\pi r^2 \lambda}{\rho c x}}$$

$$\frac{T_{v1} - T_L}{T_{v0} - T_L} = \left(\frac{V_0 - \phi_v t}{V_0} \right)^{\frac{\pi r^2 \lambda}{\rho c x}}$$

$$T_{v1} - T_L = (T_{v0} - T_L) \left(\frac{V_0 - \phi_v t}{V_0} \right)^{\frac{\pi r^2 \lambda}{\rho c x}}$$

$$T_{v1} = (T_{v0} - T_L) \left(1 - \frac{\phi_v t}{V_0} \right)^{\frac{\pi r^2 \lambda}{\rho c x}} + T_L = \underline{2917^\circ \text{C}}$$

29/56

$$l = 1 \text{ m}$$

$$r_2 = 1 \text{ cm}$$

$$d = 3 \text{ mm}$$

$$\lambda = 70 \frac{\text{W}}{\text{mK}}$$

$$\phi_m = 9,1 \text{ kg/min}$$

$$p = 1 \text{ bar}$$

$$T_n = 100^\circ \text{C}$$

$$T_2 = ?$$

$$q_i = 326 \cdot 10^6 \frac{\text{J}}{\text{kg}}$$



$$Q = \phi_m t q_i$$

$$P = S \lambda \left(-\frac{dT}{dr} \right) = l \cdot 2\pi r \lambda \left(-\frac{dT}{dr} \right) = \frac{Q}{t} = \phi_m q_i$$

$$\phi_m q_i dr = -2\pi r l \lambda dT$$

$$\int_{r_n}^{r_2} \frac{1}{r} dr = -\frac{2\pi l \lambda}{\phi_m q_i} \int_{T_n}^{T_2} dT$$

$$\ln \frac{r_2}{r_2 - d} = -\frac{2\pi l \lambda}{\phi_m q_i} (T_2 - T_n)$$

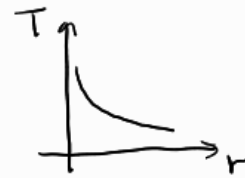
$$T_2 = T_n - \frac{\phi_m q_i}{2\pi l \lambda} \ln \frac{r_2}{r_2 - d} = 368,6 \text{ K}$$

$$\text{ce } d \ll r_2:$$

$$\ln \left(\frac{r_2}{r_2 - d} \right) = \ln \left(\frac{1}{1 - \frac{d}{r_2}} \right) = -\ln \left(1 - \frac{d}{r_2} \right)$$

$$\approx \frac{d}{r_2}$$

$$\rightarrow T_2 = T_n - \frac{\phi_m q_i d}{2\pi l \lambda}$$



12.6.3

$$r_1 = 1 \text{ cm}$$

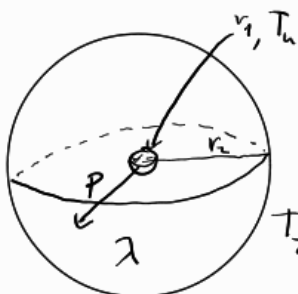
$$r_2 = 10 \text{ cm}$$

$$P = 1000 \text{ W}$$

$$T_2 = 0^\circ \text{C}$$

$$\lambda = 390 \frac{\text{W}}{\text{mK}}$$

$$T_n = ?$$



$$P = \lambda S \left(-\frac{dT}{dr} \right)$$

$$\int_{r_1}^{r_2} \frac{dr}{r^2} = -\frac{4\pi \lambda}{P} \int_{T_n}^{T_2} dT$$

$$-\frac{1}{r} \Big|_{r_1}^{r_2} = -\frac{4\pi \lambda}{P} (T_2 - T_n)$$

$$\frac{1}{r_1} - \frac{1}{r_2} = \frac{4\pi \lambda}{P} (T_n - T_2)$$

$$S = 4\pi r^2$$

$$T_n = T_2 + \frac{P \left(\frac{1}{r_1} - \frac{1}{r_2} \right)}{4\pi \lambda}$$

$$\boxed{T_n = 291,4 \text{ K}}$$

29/58 (debelina ledny, ki se nabere na vodi)

$$T_v = 0^\circ\text{C}$$

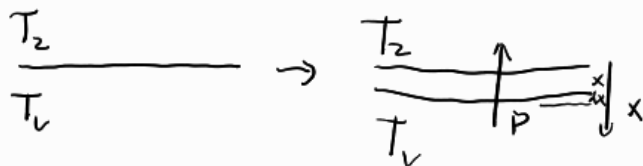
$$T_z = -10^\circ\text{C}$$

$$t = 12\text{h}$$

$$\lambda = 2,2 \frac{\text{W}}{\text{mK}}$$

$$\rho = 0,9 \frac{\text{g}}{\text{cm}^3}$$

$$d = ?$$



$$P = \lambda S \left(-\frac{dT}{dx} \right) = \frac{dQ}{dt}$$

$$dQ = dm \cdot q_t$$

$$dm = \rho dV = \rho S dx$$

$$P = \lambda S \left(\frac{\Delta T}{x} \right) = \frac{\rho S dx q_t}{dt}$$

$$\int_0^d x dx = \frac{\lambda \Delta T}{\rho q_t} \int_0^t dt$$

$$\frac{d^2}{2} = \frac{\lambda \Delta T}{\rho q_t} t$$

$$q_t = 3,36 \cdot 10^5 \frac{\text{J}}{\text{kg}}$$

$$d = \sqrt{\frac{2 \lambda \Delta T t}{\rho q_t}} = \underline{\underline{8\text{cm}}}$$

29/20

$$m = 1\text{kg}$$

$$T = \text{konst.}$$

$$p_1 = 1\text{bar}$$

$$p_2 = 50\text{bar}$$

$$\chi = 5 \cdot 10^{-5} \text{bar}^{-1}$$

$$A = ?$$

$$dA = -p dV$$

$$dA = \chi V(p) p dp$$

$$A = \chi V \int_{p_1}^{p_2} p dp =$$

$$= \chi V \frac{p_2^2 - p_1^2}{2} = 6,25\text{J}$$

$$\frac{dV}{V} = -\chi dp \rightarrow$$

$$dV = -\chi V dp$$

$$\int_{V_1}^V \frac{dV}{V} = -\int_{p_1}^p \chi dp$$

$$\ln \frac{V}{V_1} = -\chi (p - p_1)$$

$$V = V_1 e^{-\chi (p - p_1)}$$

(29/21)

$$m = 1 \text{ kg}$$

$$p_1 = 1 \text{ bar}$$

$$T_1 = 19^\circ \text{C}$$

$$T_2 = 25^\circ \text{C}$$

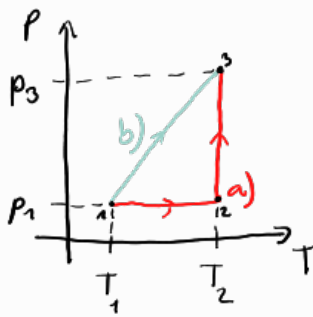
$$\chi = 5 \cdot 10^{-5} \text{ bar}^{-1}$$

$$\beta = 2 \cdot 10^{-4} \text{ K}^{-1}$$

$$p_3 = 35 \text{ bar}$$

$$a) p_{1 \rightarrow 2} = \text{konst}$$

$$T_{2 \rightarrow 3} = \text{konst}$$



$$a) A = - \int p dV$$

$$\frac{dV}{V} = \beta dT - \chi dp$$

$$1 \rightarrow 2:$$

$$\frac{dV}{V} = \beta dT$$

$$\ln \frac{V}{V_1} = \beta (T - T_1) \quad A_{12} = - \int_{p_1}^{p_2} V \beta dT =$$

$$V = V_1 e^{\beta(T-T_1)} \underset{\approx 1}{=} -p_1 \beta \int_{T_1}^{T_2} V dT =$$

$$\underset{\approx 1}{=} -p_1 \beta V \int_{T_1}^{T_2} dT =$$

$$\approx -0,12 \text{ J}$$

$$2 \rightarrow 3:$$

$$\frac{dV}{V} = -\chi dp$$

$$V = V_2 e^{-\chi \Delta p} = V_2$$

$$A_{23} = \int \chi V p dp =$$

$$= \chi V \frac{p_3^2 - p_1^2}{2} = 3,1 \text{ J}$$

$$A \approx 3 \text{ J}$$

$$b) p \propto T$$

$$A = ?$$

$$b) \frac{dV}{V} = \beta dT - \chi dp$$

$$= dT(\beta - k\chi)$$

$$p = \frac{p_3 - p_1}{T_2 - T_1} T + h$$

$$p_1 = k T_1 + h$$

$$h = p_1 - k T_1 \rightarrow p = p_1 + k(T - T_1)$$

$$dp = k dT$$

$$V = V_1 e^{(\beta - k\chi) \Delta T}$$

$$dA = -p dV =$$

$$= -p(\beta - k\chi) dT$$

$$A = - \int (k(T - T_1) + p_1) V_1 (\beta - k\chi) dT =$$

$$= (k\chi - \beta) V_1 \int_{T_1}^{T_2} (kT + (p_1 - kT_1)) dT =$$

$$= (k\chi - \beta) V_1 \left(k \frac{T_2^2 - T_1^2}{2} + (p_1 - kT_1)(T_2 - T_1) \right) =$$

$$= (k\chi - \beta) V_1 \left(\frac{(p_3 - p_1)(T_2 + T_1)}{2} + p_1(T_2 - T_1) - T_1(p_3 - p_1) \right) =$$

$$= (k\chi - \beta) V_1 \left((p_3 - p_1) \frac{T_2 + T_1 - 2T_1}{2} + p_1(T_2 - T_1) \right) =$$

$$= (k\chi - \beta) V_1 \left(\frac{T_2 - T_1}{2} (p_3 - p_1 + 2p_1) \right) = (k\chi - \beta) V_1 \frac{(T_2 - T_1)(p_3 + p_1)}{2} = \underline{\underline{0,89 \text{ J}}}$$

$$m = 1 \text{ kg} \quad c_v = 4160 \frac{\text{J}}{\text{kgK}}$$

$$T = 20^\circ\text{C} \quad \beta = 2 \cdot 10^{-4} \text{ K}^{-1}$$

$$p = 1 \text{ bar} \quad \chi = 5,6 \cdot 10^{-5} \text{ bar}^{-1}$$

$$V = \text{konst.}$$

$$dW_h = dQ = m c_v dT$$

$$H = U_h + pV$$

$$dH = m c_v dT - p dV + d(pV) = m c_v dT + d(pV) =$$

$$\frac{dU}{V} = \beta dT - \chi dp = 0$$

$$= m c_v dT + V \frac{\beta}{\chi} dT$$

$$\Delta H = \int_T^{T_{\text{isoT}}} (m c_v + V \frac{\beta}{\chi}) dT = (m c_v + V \frac{\beta}{\chi}) \Delta T$$

$$\beta dT = \chi dp$$

Σ 9/24

$$M = 46 \text{ kg/kmol}$$

$$T_v = 200^\circ\text{C}$$

$$m = 1 \text{ kg}$$

$$p = \text{konst.}$$

$$\Delta W_h = 4,02 \cdot 10^5 \text{ J}$$

$$q_i = ?$$

$$\Delta W_h = Q + A = m q_i - p \Delta V$$

$$q_i = \frac{\Delta W_h + \frac{m}{M} RT}{m} =$$

$$= \frac{\Delta W_h}{m} + \frac{R}{M} T = 4,88 \cdot 10^5 \frac{\text{J}}{\text{kg}}$$

$$p_0 V = \frac{m}{M} RT - p V_0$$

$$V_0 = \frac{m}{\rho} < 1 \text{ l}$$

↓
zun.

12.5/9

Vodka

$$m = 2 \text{ kg}$$

$$T = -6^\circ\text{C}$$

$$?$$

$$\textcircled{1} T: -6^\circ\text{C} \rightarrow 0^\circ\text{C}$$

$$Q_1 = m c \Delta T = 2 \text{ kg} \cdot 4200 \frac{\text{J}}{\text{kgK}} \cdot 6 \text{ K} = 50,4 \text{ kJ}$$

$$\textcircled{2} Q_1 \rightarrow \text{led}$$

$$Q_1 = m_2 q_i$$

$$m_2 = \frac{50,4 \text{ kJ}}{336 \text{ kJ}} = 0,15 \text{ kg}$$

$$\rightarrow \text{dabino } 0,15 \text{ kg ledu in } 1,85 \text{ kg vode pri } T = 0^\circ\text{C}$$

12.2/1

He

$$p(V) = \frac{C}{\sqrt{V}}$$

$$C = 2 \text{ bar}\sqrt{\text{J}} \quad p_2 = 2 \text{ bar}$$

$$p_1 = 1 \text{ bar} \quad V_2 = 1 \text{ l}$$

$$V_1 = 4 \text{ l} \quad C_v = \frac{3}{2} R$$

$$A = ?$$

$$\Delta W_h = ?$$

$$Q = ?$$

$$\frac{\Delta S}{m} = ?$$



$$A = -\int p dV = -C \int V^{-\frac{1}{2}} dV = -2C\sqrt{V} \Big|_{V_1}^{V_2} =$$

$$= 2C(\sqrt{V_1} - \sqrt{V_2}) = 400 \text{ J}$$

$$\Delta W_h = m c_v \Delta T =$$

$$= m \frac{3}{2} R \cdot \frac{C M}{M R} (\sqrt{V_2} - \sqrt{V_1}) =$$

$$= -300 \text{ J}$$

$$pV = \frac{m}{M} RT$$

$$T = C \cdot \sqrt{V} \frac{M}{m R}$$

$$Q = \Delta W_h - A = -700 \text{ J}$$

$$\Delta S = \int_0^Q \frac{dQ}{T} = \int_1^2 \frac{dW_h - dA}{T} = m c_v \int_{T_1}^{T_2} \frac{dT}{T} + \int_{V_1}^{V_2} \frac{p dV}{T} = m c_v \ln \frac{T_2}{T_1} + \int \frac{C dV m R}{\sqrt{V} M R \sqrt{V} V} =$$

$$= m c_v \ln \frac{\sqrt{V_2}}{\sqrt{V_1}} + \frac{m R}{n} \ln \frac{V_2}{V_1} = \frac{3}{4} m \frac{R}{n} \ln \frac{V_2}{V_1} + m \frac{R}{n} \ln \frac{V_2}{V_1} =$$

$$= \frac{7}{4} m \frac{R}{n} \ln \frac{V_2}{V_1}$$

$$\rightarrow \boxed{\frac{\Delta S}{m} = \frac{7}{4} \frac{R}{n} \ln \frac{V_2}{V_1}}$$

12.2/2

$$O_2 \quad c_v = \frac{5}{2} \frac{R}{n}$$

$$T_1 = 0^\circ C$$

$$V_1 = 3l$$

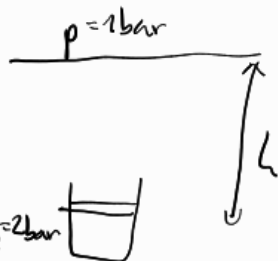
$$h = 10m$$

$$V_2 = 4l$$

$$A = ?$$

$$Q = ?$$

$$\Delta S = ?$$



$$p_2 = p_1$$

$$pV = \frac{m}{n} RT$$

$$m = \frac{pV_1 n}{RT_1}$$

$$A = \int_{V_1}^{V_2} p dV = -p(V_2 - V_1) = -200J$$

$$Q = m c_v \Delta T - A = m c_v \frac{T_1}{V_1} (V_2 - V_1) + p(V_2 - V_1) =$$

$$\frac{T_1}{V_1} = \frac{T_2}{V_2} \quad = \frac{pV_1 n}{2n} \frac{5R}{V_1} \frac{T_1}{V_1} (V_2 - V_1) + p(V_2 - V_1) =$$

$$= \frac{7}{2} p(V_2 - V_1) = 700J$$

$$\Delta S = \int \frac{dQ}{T} = \frac{7}{2} p \frac{V_1}{T_1} \int_{V_1}^{V_2} \frac{dV}{V} =$$

$$= \frac{7}{2} p \frac{V_1}{T_1} \ln \frac{V_2}{V_1} = 2,2 \frac{J}{K}$$

$$Q(V) = \frac{7}{2} p(V - V_1) \quad (\text{adijabno})$$

$$dQ = \frac{7}{2} p dV$$

29/14

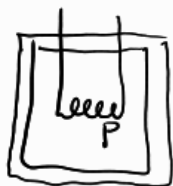
$$m = 0,5 kg$$

$$T_1 = 10^\circ C$$

$$P = 500W$$

$$t = 5min$$

$$\Delta S = ?$$



Sprememba je ireverzibilna (ne moremo iz toplote vode dobiti nazaj elektriko)

$$A = 0$$

$$\Delta W_h \approx Q = Pt = m c_v \Delta T \rightarrow \Delta T = \frac{Pt}{m c_v}$$

$$dS = \frac{dQ}{T} = \frac{m c_v dT}{T}$$

$$\rightarrow \Delta S = m c_v \ln \frac{T_2}{T_1} = m c_v \ln \left(1 + \frac{Pt}{m c_v T_1} \right) = 4,70 \frac{J}{K}$$

29/16



$$\Delta S = \Delta S_1 + \Delta S_2$$

$$dS = \int \frac{dQ}{T} = \int \frac{m c_v dT}{T} = c_v \left(m_1 \int_{T_1}^{T_k} \frac{dT}{T} + m_2 \int_{T_2}^{T_k} \frac{dT}{T} \right) = c_v \left(m_1 \ln \frac{T_k}{T_1} + m_2 \ln \frac{T_k}{T_2} \right)$$

$$m_1 \Delta T_1 = m_2 \Delta T_2$$

$$m_1 (T_1 - T_k) = m_2 (T_k - T_2)$$

$$T_k (m_1 + m_2) = T_1 m_1 + T_2 m_2$$

$$a) m_1 = 0,1 kg$$

$$T_1 = 100^\circ C$$

$$m_2 = 0,9 kg$$

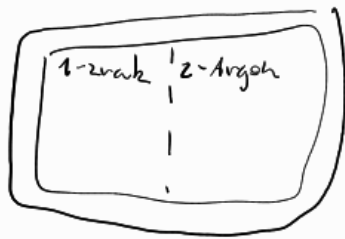
$$T_2 = 0^\circ C$$

b) $m_1 = 0,4 \text{ kg}$
 $T_1 = 80^\circ\text{C}$
 $m_2 = 0,6 \text{ kg}$
 $T_2 = 40^\circ\text{C}$

$T_k = \begin{cases} 10^\circ\text{C}; & \text{a)} \\ 56^\circ\text{C}; & \text{b)} \end{cases} \rightarrow \begin{aligned} \Delta S_a &= 20 \frac{\text{J}}{\text{K}} \\ \Delta S_b &= 7,3 \frac{\text{J}}{\text{K}} \end{aligned}$

$\Delta S_a, \Delta S_b$

12.2/5



$V_1 = 3 \text{ l}$ $M_1 = 29 \frac{\text{kg}}{\text{kmol}}$
 $m_1 = 10 \text{ g}$ $\gamma_1 = 1,4$
 $T_1 = 0^\circ\text{C}$ $c_{v1} = 715 \frac{\text{J}}{\text{kgK}}$
 $V_2 = 5 \text{ l}$ $M_2 = 40 \frac{\text{kg}}{\text{kmol}}$
 $m_2 = 20 \text{ g}$ $\gamma_2 = 1,67$
 $T_2 = 50^\circ\text{C}$ $c_{v2} = 310 \frac{\text{J}}{\text{kgK}}$
 $T_k = ?$
 $\Delta S = ?$

$$T_k = \frac{m_1 c_{v1} T_1 + m_2 c_{v2} T_2}{m_1 c_{v1} + m_2 c_{v2}} = 296,2 \text{ K}$$

$$\Delta S_1 = m_1 c_{v1} \ln \frac{T_k}{T_1} + \frac{m_1 R}{n_1} \ln \frac{V_1 + V_2}{V_1} = 3,4 \frac{\text{J}}{\text{K}}$$

$$\Delta S_2 = 1,4 \frac{\text{J}}{\text{K}}$$

29/28

$dQ = 0$
 $m = 1 \text{ kg}$
 He
 $T_1 = 90^\circ\text{C}$
 $p_2 = 5 \text{ bar}$
 $T_2 = 120^\circ\text{C}$
 $M = 4 \frac{\text{kg}}{\text{kmol}}$
 $\gamma = 1,67$
 $A = ?$
 $\Delta H = ?$

1. NAČIN

$$\Delta W_h = A = m c_v \Delta T$$

$$= \frac{m}{n} \frac{R}{\gamma - 1} \Delta T = 93 \text{ kJ}$$

$$\Delta H = m c_p \Delta T = \gamma m c_v \Delta T = \gamma \Delta W_h$$

$$= 155,3 \text{ kJ}$$

2. NAČIN

$$A = \int -p dV = \frac{-1}{1-\gamma} \int p \frac{\gamma}{p} dT \frac{m R T}{M p} =$$

$$= \frac{1}{\gamma - 1} \frac{m}{n} R \Delta T$$

$$\gamma = \frac{c_p}{c_v} \rightarrow c_p = \gamma c_v$$

$$c_p - c_v = \frac{R}{n}$$

$$c_v (\gamma - 1) = \frac{R}{n}$$

$$c_v = \frac{R}{n(\gamma - 1)}$$

$$p V^\gamma = p_1 V_1^\gamma$$

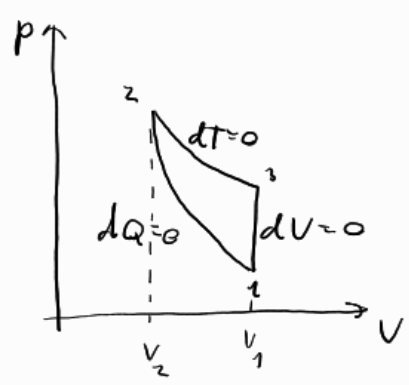
$$\frac{T}{T_1} = \left(\frac{V_1}{V} \right)^{\gamma - 1} = \left(\frac{V}{V_1} \right)^{1 - \gamma}$$

$$\ln T - \ln T_1 = (1 - \gamma) (\ln V - \ln V_1)$$

$$\frac{1}{T} dT = (1 - \gamma) \frac{1}{V} dV$$

29/49

$T_1 = 20^\circ\text{C}$
 $T_2 = 300^\circ\text{C}$
 $\eta = ?$



$\eta = \frac{|A_{\text{net}}|}{Q_{\text{dov}}}$

$(1 \rightarrow 2) \quad \Delta W_{12} = Q_{12} + A_{12} = A_{12} = m c_v (T_2 - T_1) > 0$
 $\leftarrow \begin{matrix} \text{plin} \\ \text{prejine} \end{matrix}$

$p_1 V_1^\gamma = p_2 V_2^\gamma$

$T_1 V_1^{\gamma-1} = T_2 V_2^{\gamma-1}$

$\left(\frac{V_1}{V_2}\right)^{\gamma-1} = \frac{T_2}{T_1}$

$\frac{V_1}{V_2} = \left(\frac{T_2}{T_1}\right)^{\frac{1}{\gamma-1}}$

$(2 \rightarrow 3) \quad \Delta W_{23} = Q_{23} + A_{23} = 0$

$A_{23} = - \int_2^3 p dV =$

$= - p_2 V_2 \int_{V_2}^{V_1} \frac{1}{V} dV = - p_2 V_2 \ln \frac{V_1}{V_2}$

$p_2 V_2 = p_3 V_3 = pV$

$p = p_2 \frac{V_2}{V}$

$V_1 = V_3$

$p_2 V_2 = \frac{m}{\gamma} R T_2$

$= - \frac{m}{\gamma} R T_2 \ln \left(\frac{T_2}{T_1}\right)^{\frac{1}{\gamma-1}} =$

$= - \frac{m}{\gamma} \frac{R T_2}{\gamma-1} \ln \frac{T_2}{T_1} < 0$
 $\leftarrow \text{odda}$

$Q_{23} = \frac{m}{\gamma} \frac{R T_2}{\gamma-1} \ln \frac{T_2}{T_1} > 0$
 $\leftarrow \text{prejine}$

$(3 \rightarrow 1) \quad \Delta W_{31} = Q_{31} + A_{31} = Q_{31} = m c_v (T_1 - T_2) < 0$
 $\leftarrow \text{odda}$

$\eta = \frac{|A_{\text{net}}|}{Q_{23}} = \frac{\frac{m}{\gamma} \frac{R T_2}{\gamma-1} \ln \frac{T_2}{T_1} - m c_v (T_2 - T_1) \frac{R}{\gamma-1}}{\frac{m}{\gamma} \frac{R T_2}{\gamma-1} \ln \frac{T_2}{T_1}} =$

$= \frac{T_2 \ln \frac{T_2}{T_1} - (T_2 - T_1)}{T_2 \ln \frac{T_2}{T_1}} = 1 - \frac{T_2 - T_1}{T_2 \ln \frac{T_2}{T_1}} = \underline{\underline{0.27}}$

Hitroje za Q_{23} :

$\Delta S = 0 = S_{23} + S_{31}$

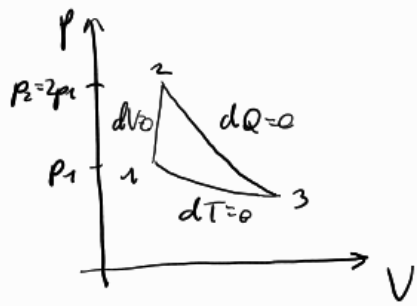
$S_{23} = \int \frac{dQ}{T} = \frac{Q_{23}}{T_2}$

$S_{31} = \int \frac{dQ}{T} = m c_v \ln \frac{T_1}{T_2}$

$Q_{23} = -T_2 m c_v \ln \frac{T_1}{T_2} = T_2 m c_v \ln \frac{T_2}{T_1} = -A_{23}$

12.4.3

$m = 1 \text{ kg}$
 $M = 58 \frac{\text{kg}}{\text{kmol}}$
 $T_1 = 0^\circ\text{C}$
 $C_v = 360 \frac{\text{J}}{\text{kg K}}$



$\gamma = 1,4$
 $\eta = ?$

(1→2) $\Delta W_{12} = A_{12}^0 + Q_{12} = m C_v (T_2 - T_1) = m C_v T_1 \left(\frac{P_2}{P_1} - 1 \right) = m C_v T_1$
 $\frac{P_1}{T_1} = \frac{P_2}{T_2} \rightarrow T_2 = T_1 \left(\frac{P_2}{P_1} \right)$
adversario

(2→3) $\Delta W_{23} = A_{23}^0 + Q_{23} = m C_v (T_3 - T_2) = m C_v T_1 (1 - 2) = -m C_v T_1 < 0$
adversario

(3→1) $A_{31} = -Q_{31}$
 $\Delta S = 0 = S_{31} + S_{12}$

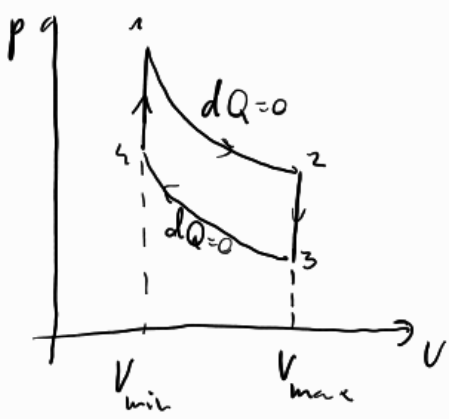
$\eta = \frac{|A_{23}| - |A_{31}|}{Q_{12}} =$
 $= \frac{m C_v T_1 (1 - \ln 2)}{m C_v T_1} = 1 - \ln 2 = \underline{\underline{0,31}}$

$S_{12} = \int_1^2 \frac{dQ}{T} = m C_v \int_1^2 \frac{dT}{T} = m C_v \ln \frac{T_2}{T_1} =$
 $= m C_v \ln \frac{T_1 \frac{P_2}{P_1}}{T_1} = m C_v \ln \frac{P_2}{P_1}$
 $S_{31} = \frac{Q_{31}}{T_1}$

$Q_{31} = -m C_v T_1 \ln \frac{P_2}{P_1} < 0$
 $A_{31} = m C_v T_1 \ln \frac{P_2}{P_1} > 0$
adversario
dovedeno

12.4.4

$\gamma = 1,4$
 $\frac{V_{max}}{V_{min}} = 8$
 $\eta = ?$
 $\Delta S = ?$



$\Delta S = 0$

(1→2) $\Delta W_{12} = A_{12} = m C_v (T_2 - T_1) > 0$
dodano

(2→3) $\Delta W_{23} = Q_{23} = m C_v (T_3 - T_2) < 0$
adversario

$$(3 \rightarrow 4) \quad \Delta W_{n34} = A_{34} = m c_v (T_4 - T_3) < 0 \quad \text{oddano}$$

$$(4 \rightarrow 1) \quad \Delta W_{n41} = Q_{41} = m c_v (T_1 - T_4) > 0 \quad \text{dodano}$$

$$V_{\min} = V_1 = V_4$$

$$V_{\max} = V_2 = V_3$$

$$T_1 V_1^{\gamma-1} = T_2 V_2^{\gamma-1}$$

$$T_1 = T_2 \left(\frac{V_{\max}}{V_{\min}} \right)^{\gamma-1}$$

$$T_3 V_3^{\gamma-1} = T_4 V_4^{\gamma-1}$$

$$T_4 = T_3 \left(\frac{V_{\max}}{V_{\min}} \right)^{\gamma-1}$$

$$\Delta S = 0 = S_{23} + S_{41}$$

$$S_{23} = -S_{41}$$

$$n R \ln \frac{T_3}{T_2} = -n R \ln \frac{T_4}{T_1} = n R \ln \frac{T_4}{T_1}$$

$$\frac{T_3}{T_2} = \frac{T_4}{T_1}$$

$$\eta = \frac{|A_{34}| - |A_{41}|}{Q_{41}} = \frac{(T_3 - T_4) - (T_2 - T_1)}{T_1 - T_4} =$$

$$= \frac{T_1 - T_4 + T_3 - T_2}{T_1 - T_4} = 1 - \frac{T_2 - T_3}{T_1 - T_4} =$$

$$= 1 - \frac{T_2 \left(1 - \frac{T_3}{T_2}\right)}{T_1 \left(1 - \frac{T_4}{T_1}\right)} = 1 - \frac{1}{8^{\gamma-1}} = \underline{\underline{0.56}}$$

29/51

kladilnik

$$\eta P = \frac{dQ}{dt} = \frac{dm q_t}{dt} = \phi_m q_t$$

$$\eta_c = \frac{T_{\min}}{T_{\max} - T_{\min}}$$

$$\phi_m = \frac{1}{10} \frac{\text{kg}}{\text{min}}$$

$$T_{\min} = 0^\circ \text{C}$$

$$T_{\max} = 30^\circ \text{C}$$

$$\eta = \frac{\eta_c}{2.5}$$

$$P = ?$$

$$Q_t = 3.36 \cdot 10^5 \frac{\text{J}}{\text{kg}}$$

$$P = \phi_m q_t \cdot 2.5 \frac{T_{\max} - T_{\min}}{T_{\min}} = \frac{1 \cdot \text{kg} \cdot 3.36 \cdot 10^5 \cdot 2.5 \cdot 30 \text{K}}{10 \cdot 60 \cdot 5 \cdot \text{kg} \cdot 273 \text{K}} = \underline{\underline{153 \text{W}}}$$

12.4.5 Hladilnik

$$m = 10 \text{ kg}$$

$$T_0 = 20^\circ\text{C}$$

$$T_1 = -193^\circ\text{C}$$

$$c_p = 1000 \frac{\text{J}}{\text{kgK}}$$

$$A = ?$$

↓
kakšno naj deluje, da bo $A = \min$

$$\eta = \frac{Q_{\min}}{A} = \frac{T_{\min}}{T_{\max} - T_{\min}} \rightarrow \eta(T_{\min}) = \frac{1}{\frac{T_{\max}}{T_{\min}} - 1}$$

$T_{\min}$ se spreminja

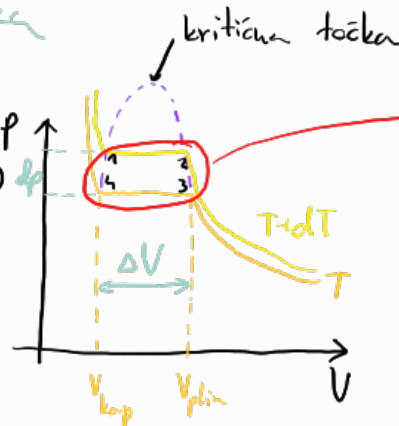
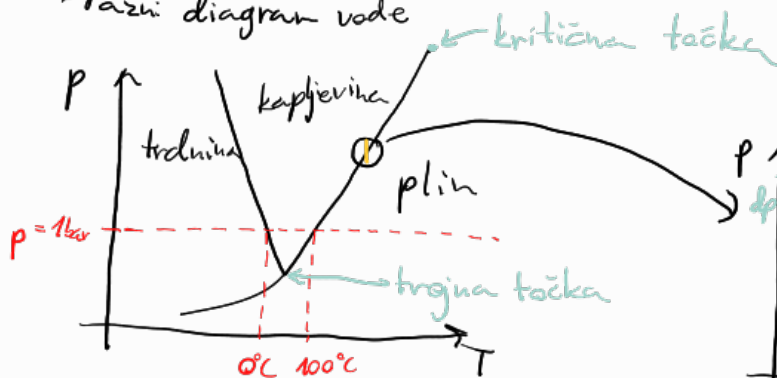
$$dA = \frac{dQ_{\min}}{\eta} = \left(\frac{T_{\max}}{T_{\min}} - 1 \right) m c_p (-dT_{\min})$$

$$A = m c_p \int_{T_0}^{T_1} \left(1 - \frac{T_{\max}}{T_{\min}} \right) dT_{\min} = m c_p \left((T_1 - T_0) - T_0 \ln \frac{T_1}{T_0} \right) =$$

$$= m c_p \left(T_1 + T_0 \left(\ln \frac{T_0}{T_1} - 1 \right) \right) = 1,76 \cdot 10^6 \text{ J}$$

CLAUS-CLAPEYRON

- Fazni diagram vode



Carnot krožna spr.

$$\Delta S = 0 = S_{12} + 0 + S_{34} + 0 =$$

$$= \frac{Q_{12}}{T+dT} + \frac{Q_{34}}{T}$$

$$-Q_{34} = Q_{12} + dA$$

$$dA = -\Delta V dp$$

$$\frac{Q_{12}}{T+dT} = \frac{Q_{12} + dA}{T}$$

$$Q_{12} T = Q_{12} T + Q_{12} dT + dA T + dT dA$$

$$Q_{12} dT = -dA \cdot T$$

$$Q_{12} dT = \Delta V T dp$$

$$\frac{dp}{dT} = \frac{Q_{12}}{\Delta V T} = \frac{L}{\Delta V T}$$

latentna toplota

$$L = \begin{cases} m q_i \\ m q_c \end{cases}$$

12.3.3

$$p_0 = 1 \text{ bar}$$

$$T_0 = 20^\circ\text{C}$$

$$V_0 = 1 \text{ m}^3$$

$$m_p = 0,99$$

$$p_p = ?$$

$$p_2 = ?$$

$$\eta = ?$$

$$m_2 = ?$$

$$p_1 + p_2 = p_0 \quad p_1 = \frac{m_p}{n_p} \frac{R T_0}{V_0} = 122 \text{ Pa}$$

$$p_2 = 99878 \text{ Pa} \quad m_2 = \frac{M_2 p_2 V_0}{R T_0} = 1,189 \text{ kg}$$

$$\frac{dp}{dT} = \frac{L}{\Delta V T} = \frac{q_i}{\left(\frac{M}{\rho} + \frac{R T}{p} \right) T} \approx \frac{q_i}{\frac{R T^2}{p}} = \frac{q_i p}{R T^2}$$

$$\int_{p_0}^p \frac{dp}{p} = \frac{q_i}{R} \int_{T_0}^T \frac{dT}{T^2} \Rightarrow \ln \frac{p}{p_0} = -\frac{q_i}{R} \left(\frac{1}{T} - \frac{1}{T_0} \right)$$

$$p = p_0 e^{-\frac{q_i}{R} \left(\frac{1}{T} - \frac{1}{T_0} \right)} \Rightarrow p_{\max} = p_0 e^{-\frac{q_i}{R} \left(\frac{1}{293} - \frac{1}{273} \right)} = 2,6 \cdot 10^3 \text{ Pa}$$

$$\eta = \frac{p_1}{p_2} = \frac{p_p}{p_{\max}} = \frac{122 \text{ Pa}}{2,6 \cdot 10^3 \text{ Pa}} = 4,67\%$$

$$p_1 V_0 = p_2 V_2 = p_{\max} V_2 \rightarrow V_2 = \frac{p_1}{p_{\max}} V_0 = \eta V_0 = 46,7 \text{ l}$$

$$p_2 = p_{\max} = 2,6 \cdot 10^3 \text{ Pa}$$

$$V_2 = ?, p_2 = ?$$